

Total Marks: 7.5

Obtained Marks: _____

Object Oriented Programming Lab

Final project

Inventory Management System
(Used by businesses to manage stock)

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1. Executive Summary

The Intelligent Inventory Management System (IIMS) is a robust software solution developed using C++ and Object-Oriented Programming principles to streamline inventory operations within business environments. The system provides automated stock management, sales processing, secure authentication, and persistent data storage through a modular architecture.

By integrating encapsulation, inheritance, polymorphism, and abstraction, the project demonstrates the practical application of modern software engineering practices while addressing real-world inventory management challenges. The resulting system is scalable, maintainable, and adaptable for future enhancements.

2. Objectives of the Project

The scope of the Intelligent Inventory Management System includes:

- ⑩ Product registration and management
- ⑩ Inventory monitoring and stock updates
- ⑩ Sales transaction processing
- ⑩ User authentication and authorization
- ⑩ Data persistence using file handling
- ⑩ Reporting and transaction history

The system is intended for small and medium-sized businesses requiring a lightweight inventory management solution.

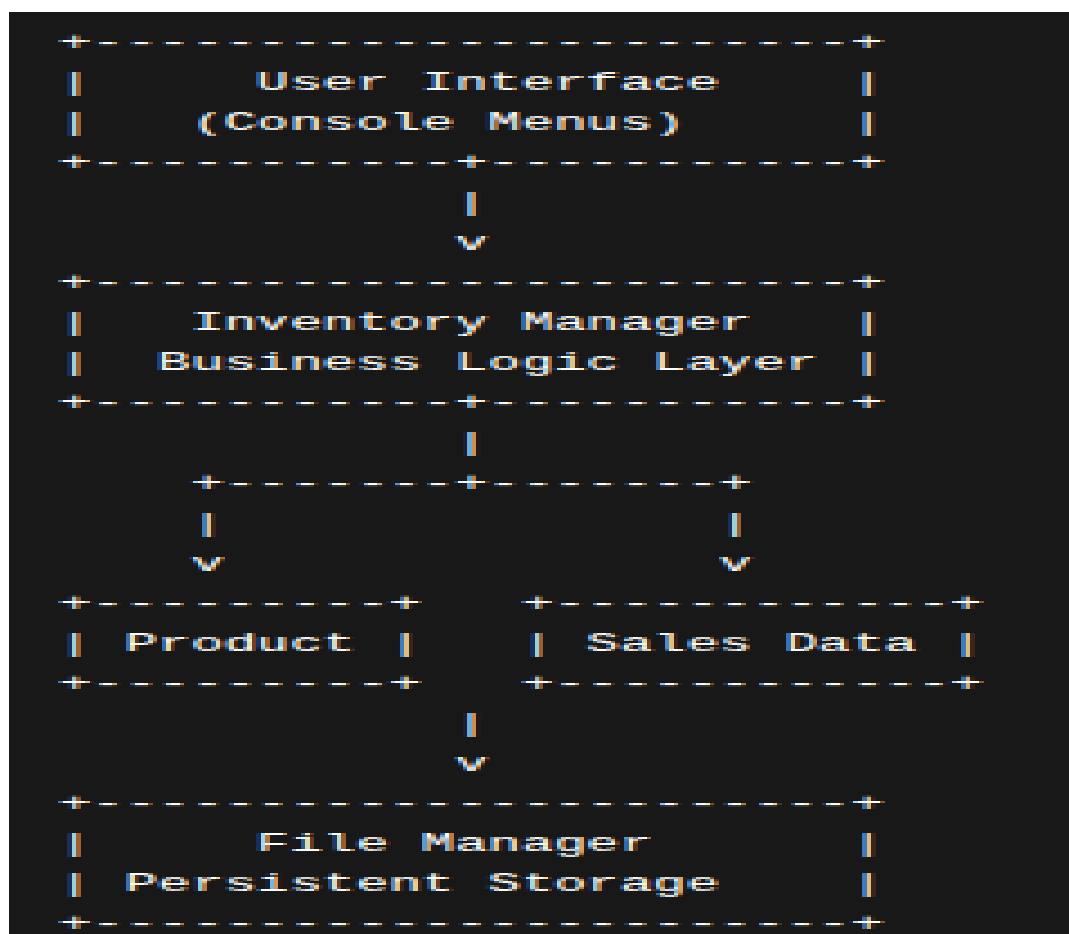
3. Functional Features

- ☐ 1. Add New Products – Register product details (ID, name, price, quantity).
- ☐ 2. Update Inventory – Modify stock quantities.
- ☐ 3. Search Products – Retrieve product information using ID or name.
- ☐ 4. Record Sales – Deduct sold quantities and calculate total price.
- ☐ 5. View Inventory and Sales – Display current stock and sales history.
- ☐ 6. Role-Based Access Control – Admin/User login for secure operations.
- ☐ 7. Persistent Data Storage – Store data in files for long-term use.

4. Core OOP Concepts Applied

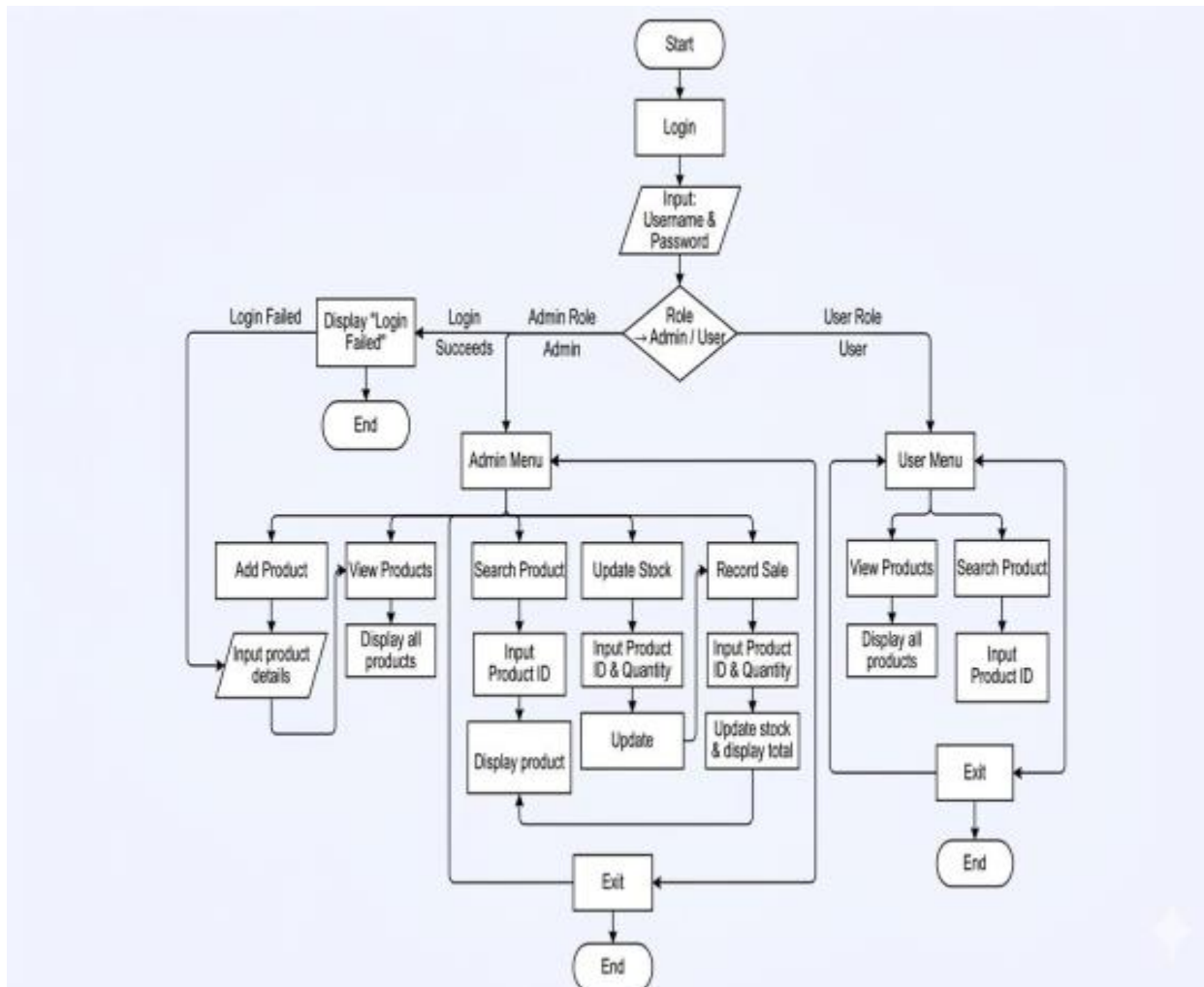
Concept	How It's Used in Inventory Management System
Encapsulation	Product details like price, ID, and quantity are kept private inside the Product class. You can only access or change them through proper getter and setter functions.
Abstraction	From the user's side, it's just a simple menu. All the behind-the-scenes logic — stock deductions, total calculations — stays hidden inside the class methods.
Inheritance	A base User class holds shared attributes like username and password. Admin and Customer are derived from it, each with their own specific permissions.
Polymorphism	The << operator is overloaded for clean product output, and virtual functions allow the system to handle different product types like perishable and non-perishable items.

5. System Architecture & Features



6. Flow Chart

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6.1 Explanation Of Flowchart

- ⑩ Upon successful login, the Admin gains full system privileges — adding products, updating stock, and recording sales.
- ⑩ A secure login authenticates users into either Admin or Standard User roles.
- ⑩ The Standard User is limited to viewing and searching product data only.
- ⑩ Each action leads to targeted input prompts such as Product ID or Quantity, keeping data modification guided and structured.
- ⑩ This architecture reinforces the OOP principle of Encapsulation by restricting critical inventory modifications to authorized administrative paths only.
- ⑩ Both user types conclude with a formal exit point, maintaining a clean and professional program termination.

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7. Future Enhancements

- ☐ 1. Graphical User Interface (GUI) – Enhance usability.
- ☐ 2. Low Stock Alerts – Automatic notifications for low inventory.
- ☐ 3. Supplier Management – Track supplier details and automate reordering.
- ☐ 4. Detailed Reporting – Generate analytical reports for sales and inventory.
- ☐ 5. Database Integration – Use SQL or NoSQL databases for large-scale applications.
- ☐ 6. Cloud Connectivity – Enable multi-location inventory management.

8. System Design

- ☐ 1. Design class structures for Product and Inventory to encapsulate data.
- ☐ 2. Decide on dynamic data storage using vectors.
- ☐ 3. Create a menu-driven interface for user interaction.
- ☐ 4. Plan optional enhancements such as file handling and login system.

9. Source Code

```
#include <iostream>
#include <fstream>
#include <sstream>
#include <vector>
#include <string>
#include <iomanip>
#include <cstdlib>
using namespace std;

// ===== PRODUCT CLASS =====

class Product
{
private:
```

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```
int id;  
string name;  
double price;  
int quantity;
```

public:

```
Product() { id=0; price=0; quantity=0; }
```

```
Product(int i,string n,double p,int q)
```

```
{  
    id=i;  
    name=n;  
    price=p;  
    quantity=q;  
}
```

```
int getID() { return id; }
```

```
string getName() { return name; }
```

```
double getPrice() { return price; }
```

```
int getQuantity() { return quantity; }
```

```
void setQuantity(int q) { quantity=q; }
```

```
void setPrice(double p) { price=p; }
```

```
friend ostream& operator<<(ostream &out, Product &p)
```

```
{  
    out<<"ID: "<<p.id<<endl;  
    out<<"Name: "<<p.name<<endl;  
    out<<"Price: "<<p.price<<endl;
```

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```
out<<"Quantity: "<<p.quantity<<endl;

return out;

}

string toFile()
{
    stringstream ss;
    ss << id << "|" << name << "|" << price << "|" << quantity;
    return ss.str();
}

};

// ===== USER CLASSES =====

class User
{
protected:
    string username;
    string password;

public:
    User(string u,string p)
    {
        username=u;
        password=p;
    }

    virtual void menu() = 0;
};
```


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```
class Admin : public User
{
public:
    Admin(string u,string p):User(u,p){}

    void menu()
    {
        cout<<"\nLogged in as Admin\n";
    }
};

class Customer : public User
{
public:
    Customer(string u,string p):User(u,p){}

    void menu()
    {
        cout<<"\nLogged in as User\n";
    }
};

vector<Product> products;

// ===== FILE FUNCTIONS =====

void saveProducts()
{
    ofstream file("products.txt");
```

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```
for(int i=0;i<(int)products.size();i++)
{
    file<<products[i].ToFile()<<endl;
}

file.close();
}

void loadProducts()
{
    products.clear();

    ifstream file("products.txt");

    if(!file)
        return;

    string line;

    while(getline(file,line))
    {
        int p1=line.find('"');
        int p2=line.find('"',p1+1);
        int p3=line.find('"',p2+1);

        int id=atoi(line.substr(0,p1).c_str());
        string name=line.substr(p1+1,p2-p1-1);
        double price=atof(line.substr(p2+1,p3-p2-1).c_str());
        int qty=atoi(line.substr(p3+1).c_str());
```

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```
        products.push_back(Product(id,name,price,qty));
    }

    file.close();
}

void addProduct()
{
    int id,qty;
    double price;
    string name;

    cout<<"\nEnter Product ID: ";
    cin>>id;

    cin.ignore();

    cout<<"Enter Product Name: ";
    getline(cin,name);

    cout<<"Enter Price: ";
    cin>>price;

    cout<<"Enter Quantity: ";
    cin>>qty;

    products.push_back(Product(id,name,price,qty));
```

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```
saveProducts();

cout<<"\nProduct Added Successfully.\n";
}

void viewProducts()
{
    if(products.size()==0)
    {
        cout<<"\nInventory Empty.\n";
        return;
    }

    cout<<"\n===== INVENTORY =====\n";

    for(int i=0;i<(int)products.size();i++)
    {
        cout<<"\n-----\n";
        cout<<products[i];

        if(products[i].getQuantity()<5)
            cout<<"LOW STOCK ALERT!\n";
    }
}

void searchProduct()
{
    int id;
```

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```
cout<<"\nEnter Product ID: ";
cin>>id;

for(int i=0;i<(int)products.size();i++)
{
    if(products[i].getID()==id)
    {
        cout<<"\nProduct Found\n";
        cout<<products[i];
        return;
    }
}

cout<<"\nProduct Not Found.\n";
}

void updateProduct()
{
    int id;

    cout<<"\nEnter Product ID: ";
    cin>>id;

    for(int i=0;i<(int)products.size();i++)
    {
        if(products[i].getID()==id)
        {
            int qty;
            double price;
```

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```
cout<<"New Price: ";
```

```
cin>>price;
```

```
cout<<"New Quantity: ";
```

```
cin>>qty;
```

```
products[i].setPrice(price);
```

```
products[i].setQuantity(qty);
```

```
saveProducts();
```

```
cout<<"\nProduct Updated.\n";
```

```
return;
```

```
}
```

```
}
```

```
cout<<"\nProduct Not Found.\n";
```

```
}
```

```
void deleteProduct()
```

```
{
```

```
int id;
```

```
cout<<"\nEnter Product ID: ";
```

```
cin>>id;
```

```
for(int i=0;i<(int)products.size();i++)
```

```
{
```

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```
if(products[i].getID()==id)
{
    products.erase(products.begin()+i);

    saveProducts();

    cout<<"\nProduct Deleted.\n";
    return;
}

cout<<"\nProduct Not Found.\n";
}

void recordSale()
{
    int id,qty;

    cout<<"\nEnter Product ID: ";
    cin>>id;

    for(int i=0;i<(int)products.size();i++)
    {
        if(products[i].getID()==id)
        {
            cout<<"Enter Quantity Sold: ";
            cin>>qty;

            if(qty>products[i].getQuantity())
```

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```
{  
    cout<<"\nInsufficient Stock.\n";  
    return;  
}  
  
double total=qty*products[i].getPrice();  
  
products[i].setQuantity(products[i].getQuantity()-qty);  
  
saveProducts();  
  
ofstream sale("sales.txt",ios::app);  
  
sale<<"Product ID: "<<products[i].getID()  
    <<" | Name: "<<products[i].getName()  
    <<" | Qty: "<<qty  
    <<" | Total: "<<total  
    <<endl;  
  
sale.close();  
  
cout<<"\nSale Successful.\n";  
cout<<"Total Bill: "<<total<<endl;  
  
return;  
}  
}
```



```
cout<<"\nProduct Not Found.\n";
```


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```
}

void salesHistory()
{
    ifstream file("sales.txt");

    if(!file)
    {
        cout<<"\nNo Sales Found.\n";
        return;
    }

    string line;

    cout<<"\n===== SALES HISTORY =====\n";

    while(getline(file,line))
    {
        cout<<line<<endl;
    }

    file.close();
}

bool loginAdmin()
{
    string u,p;

    cout<<"\nAdmin Username: ";
```

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```
cin>>u;

cout<<"Admin Password: ";
cin>>p;

return (u=="ayaad.ullah" && p=="ayaad123");
}

bool loginUser()
{
    string u,p;

    cout<<"\nUser Username: ";
    cin>>u;

    cout<<"User Password: ";
    cin>>p;

    return (u=="azhan.khan" && p=="azhan123");
}

void adminPanel()
{
    int choice;

    do
    {
        cout<<"\n=====
```

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```
cout<<"\n INVENTORY MANAGEMENT SYSTEM OF TECHNOVA  
ELECTRONICS";
```

```
cout<<"\n=====";
```

```
cout<<"\n1. Add Product";
```

```
cout<<"\n2. View Products";
```

```
cout<<"\n3. Search Product";
```

```
cout<<"\n4. Update Product";
```

```
cout<<"\n5. Delete Product";
```

```
cout<<"\n6. Record Sale";
```

```
cout<<"\n7. Sales History";
```

```
cout<<"\n8. Save Data";
```

```
cout<<"\n9. Logout";
```

```
cout<<"\n=====";
```

```
cout<<"\nEnter Choice: ";
```

```
cin>>choice;
```

```
switch(choice)
```

```
{
```

```
    case 1: addProduct(); break;
```

```
    case 2: viewProducts(); break;
```

```
    case 3: searchProduct(); break;
```

```
    case 4: updateProduct(); break;
```

```
    case 5: deleteProduct(); break;
```

```
    case 6: recordSale(); break;
```

```
    case 7: salesHistory(); break;
```

```
    case 8: saveProducts(); cout<<"\nSaved.\n"; break;
```

```
    case 9: cout<<"\nLogging Out...\n"; break;
```

```
    default: cout<<"\nInvalid Choice.\n";
```

```
}
```

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```
} while(choice!=9);  
  
}  
  
void userPanel()  
{  
    int choice;  
  
    do  
    {  
        cout<<"\n=====";  
        cout<<"\n USER PANEL";  
        cout<<"\n=====";  
        cout<<"\n1. View Products";  
        cout<<"\n2. Search Product";  
        cout<<"\n3. Sales History";  
        cout<<"\n4. Logout";  
        cout<<"\nChoice: ";  
        cin>>choice;  
  
        switch(choice)  
        {  
            case 1: viewProducts(); break;  
            case 2: searchProduct(); break;  
            case 3: salesHistory(); break;  
            case 4: break;  
            default: cout<<"\nInvalid Choice";  
        }  
    } while(choice!=4);  
}
```

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```
int main()
{
    loadProducts();

    int choice;

    do
    {
        cout<<"\n=====";
        cout<<"\n INVENTORY MANAGEMENT SYSTEM";
        cout<<"\n=====";
        cout<<"\n1. Admin Login";
        cout<<"\n2. User Login";
        cout<<"\n3. Exit";
        cout<<"\nChoice: ";
        cin>>choice;

        switch(choice)
        {
            case 1:
                if(loginAdmin())
                {
                    Admin a("ayaad.ullah","ayaad123");
                    a.menu();
                    adminPanel();
                }
            else
                cout<<"\nInvalid Credentials.\n";
```

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```
break;

case 2:
    if(loginUser())
    {
        Customer c("azhan.khan","azhan123");
        c.menu();
        userPanel();
    }
    else
        cout<<"\nInvalid Credentials.\n";
    break;

case 3:
    saveProducts();
    cout<<"\nThank You.\n";
    break;

default:
    cout<<"\nInvalid Choice.\n";
}

} while(choice!=3);

return 0;
}
```

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10.Outputs Of Source Code:

1. Login Screen

System Login Interface

```
=====
INVENTORY MANAGEMENT SYSTEM
=====

1. Admin Login
2. User Login
3. Exit

Choice: 1

Admin Username: ayaad.ullah
Admin Password: *****
```

2. Main Menu

Administrator Dashboard

```
Logged in as Admin

=====
INVENTORY MANAGEMENT SYSTEM OF
TECHNOVA ELECTRONICS
=====

1. Add Product
2. View Products
3. Search Product
4. Update Product
5. Delete Product
6. Record Sale
7. Sales History
8. Save Data
9. Logout

=====
Enter Choice:
```

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3. Add Product Operation

Product Registration Module

```
=====
INVENTORY MANAGEMENT SYSTEM OF
TECHNOVA ELECTRONICS
=====

Enter Choice: 1

Enter Product ID: 101

Enter Product Name: Dell Laptop

Enter Price: 85000

Enter Quantity: 10

Product Added Successfully.
```

4. Inventory Display

Current Inventory Status

```
===== INVENTORY =====

-----
ID: 101
Name: Dell Laptop
Price: 85000
Quantity: 10

-----
ID: 102
Name: Gaming Mouse
Price: 3500
Quantity: 3

LOW STOCK ALERT!

-----
ID: 103
Name: Mechanical Keyboard
Price: 7500
Quantity: 15
```


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5. Sales Recording

Sales Transaction Processing

```
Enter Product ID: 101  
  
Enter Quantity Sold: 2  
  
Sale Successful.  
  
Total Bill: 170000
```

6. Sales History

Transaction History Record

```
===== SALES HISTORY =====  
  
Product ID: 101 |  
Name: Dell Laptop |  
Qty: 2 |  
Total: 170000  
  
Product ID: 102 |  
Name: Gaming Mouse |  
Qty: 1 |  
Total: 3500  
  
Product ID: 103 |  
Name: Mechanical Keyboard |  
Qty: 3 |  
Total: 22500
```

7. Product Search

```
Enter Product ID: 101  
  
Product Found  
  
ID: 101  
Name: Dell Laptop  
Price: 85000  
Quantity: 8
```

8. Update Product

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Enter Product ID: 101

New Price: 90000

New Quantity: 20

Product Updated.

11. Enhanced Future Enhancements

1. GUI Development using Qt Framework
2. SQL Database Integration
3. Barcode Scanner Support
4. Inventory Analytics Dashboard
5. Cloud Backup System
6. Multi-Branch Synchronization
7. Mobile Application Integration
8. AI-Based Demand Prediction
9. Low Stock Alert Notifications
10. Report Generation Module

12. Conclusion

The Intelligent Inventory Management System successfully demonstrates the practical implementation of Object-Oriented Programming concepts in a real-world software solution. Through the integration of inventory management, sales processing, user authentication, and persistent data storage, the project achieves all its intended objectives while maintaining scalability, maintainability, and security.

The system effectively applies the four pillars of Object-Oriented Programming and showcases professional software engineering practices including modular design, code reusability, data encapsulation, and role-based access control.

Beyond fulfilling academic requirements, the project provides a functional foundation that can be extended into a complete commercial inventory management solution through the addition of graphical interfaces, database connectivity, cloud services, and advanced analytics capabilities.

The successful completion of this project has strengthened our understanding of software development methodologies, Object-Oriented Design principles, and practical problem-solving techniques, making it a valuable learning experience and a strong demonstration of programming proficiency.